

CUAI HAR 스터디

2025.11.04

발표자 : 정현석

스터디원 소개 및 만남 인증



스터디원 1 : 권순범

스터디원 2 : 남승운

스터디원 3 : 방세현

스터디원 4 : 정현석

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스터디 컨셉

- 스터디원 간의 딥러닝 이해력 차이 有



Basic 트랙

머신러닝의 기초부터 시작하고 싶은 분들을 위해 교재를 이용한 스터디를 진행하며 매주 간단한 퀴즈를 수행합니다.
Basic Track 프로젝트를 마친 후에는 Basic과 Advanced의 구분 없이 활동하게 됩니다.

Advanced 트랙

머신러닝에 대한 경험이 있는 분들을 위한 심화 커리큘럼입니다.
한 학기 동안 본인의 관심 분야와 관련된 프로젝트와 스터디에 필수로 참여합니다.
다양한 실습을 통한 직접적인 문제 해결과 실력 향상에 중점을 두고 준비한 트랙입니다.

→ **느리더라도 천천히 제대로 함께가는 스터디!**



스터디 일정

a. 중간고사 이전

- 목표: 딥러닝 이해 수준 맞추기

- 내용: 향후 논문을 읽을 수 있게 기본 딥러닝 스터디 or
 대충 넘겨왔던 개념/논문 다시 공부

스터디 일정

b. 중간고사 이후

- 목표: HAR 맛보기

- 내용:

내용	10주차	11주차	12주차	13주차	14주차	15주차
HAR 리뷰 논문 읽기						
tutorial 돌리면서 몸으로 익히기						
HAR 다른 논문 각자 리뷰						
논문 발표						
논문 구현 (선택)						

Past, Present, and Future of Sensor-based Human Activity Recognition Using Wearables: A Surveying Tutorial on a Still Challenging Task

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SUNGHO SUH, Department of Artificial Intelligence, Korea University, Republic of Korea

PAUL LUKOWICZ, RPTU Kaiserslautern-Landau and DFKI, Germany

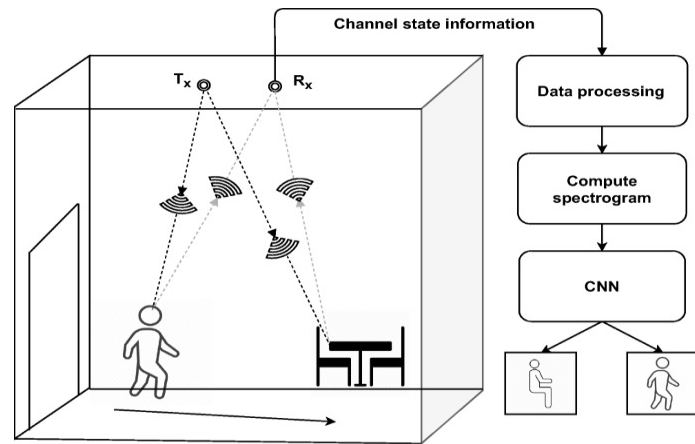
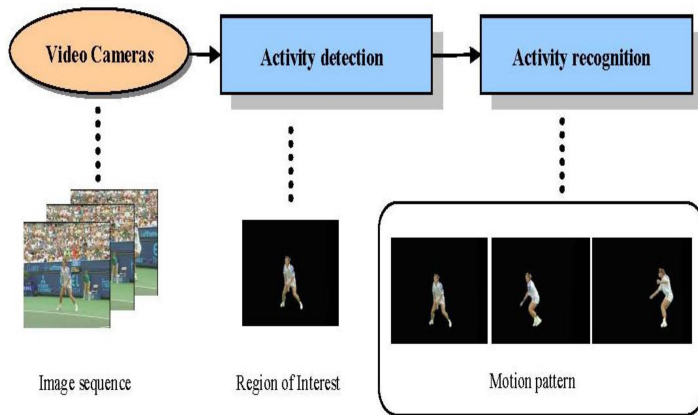
THOMAS PLOTZ, School of Interactive Computing, College of Computing, Georgia Institute of Technology, USA

In the many years since the inception of wearable sensor-based Human Activity Recognition (HAR), a wide variety of methods have been introduced and evaluated for their ability to recognize activities. Substantial gains have been made since the days of hand-crafting heuristics as features, yet, progress has seemingly stalled on many popular benchmarks, with performance falling short of what may be considered 'sufficient'—despite the increase in computational power and scale of sensor data, as well as rising complexity in techniques being employed. The HAR community approaches a new paradigm shift, this time incorporating world knowledge from foundational models. In this paper, we take stock of sensor-based HAR – surveying it from its beginnings to the current state of the field, and charting its future. This is accompanied by a hands-on tutorial, through which we guide practitioners in developing HAR systems for real-world application scenarios. We provide a compendium for novices and experts alike, of methods that aim at finally solving the activity recognition problem.

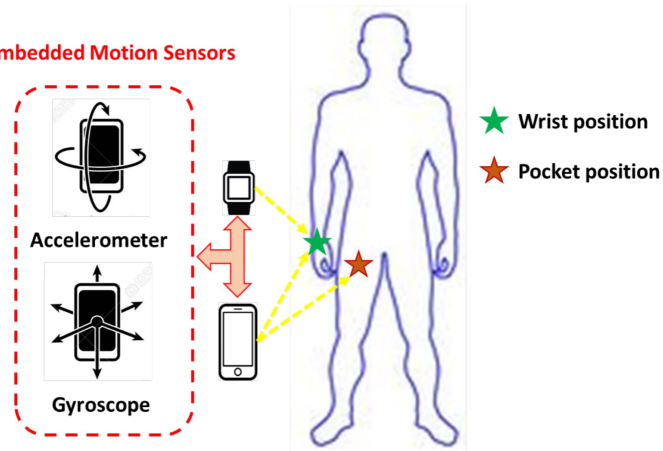
CCS Concepts: • Human-centered computing → Human computer interaction (HCI); Ubiquitous and mobile computing; • Computing methodologies → Machine learning

HAR 소개

Human Activity Recognition (HAR)

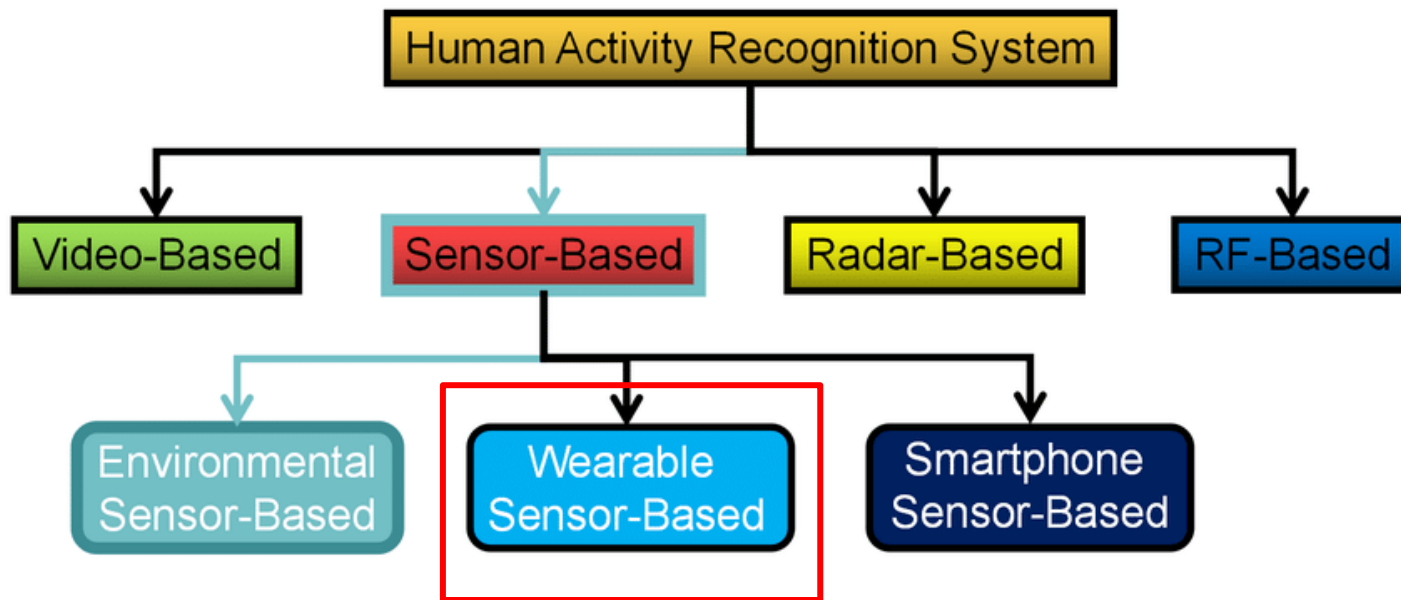


Embedded Motion Sensors



HAR 소개

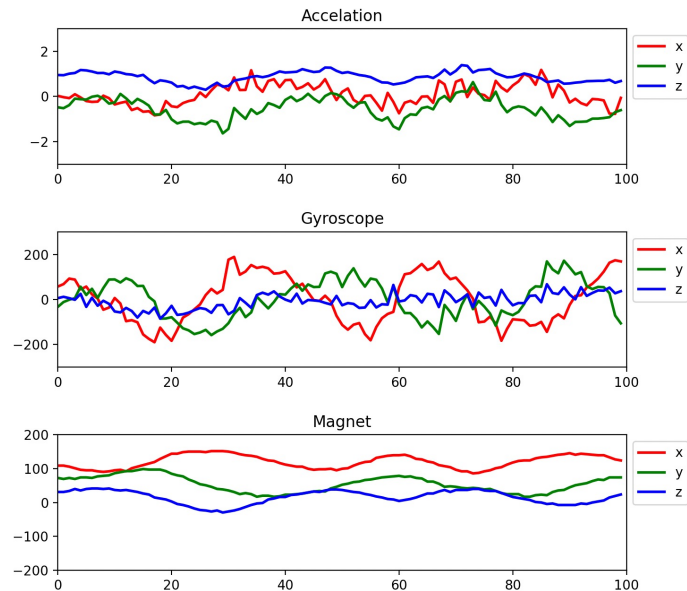
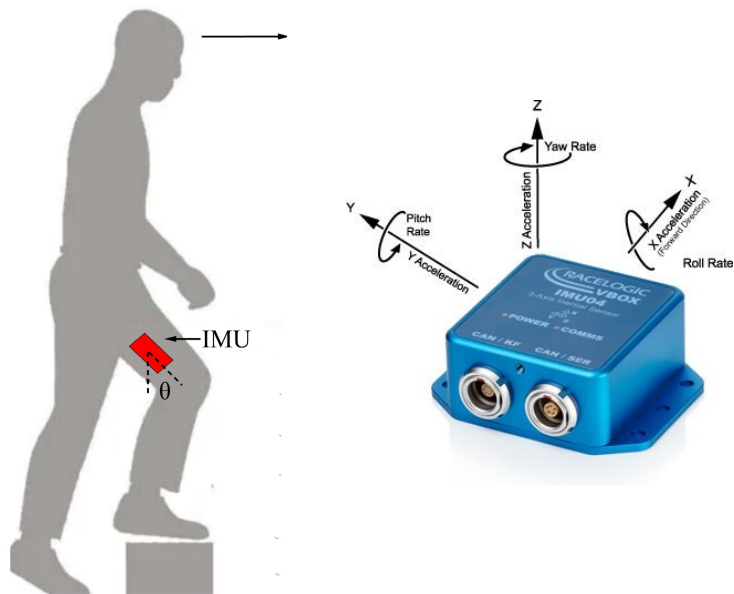
Human Activity Recognition (HAR)



이번 학기 스터디의 target

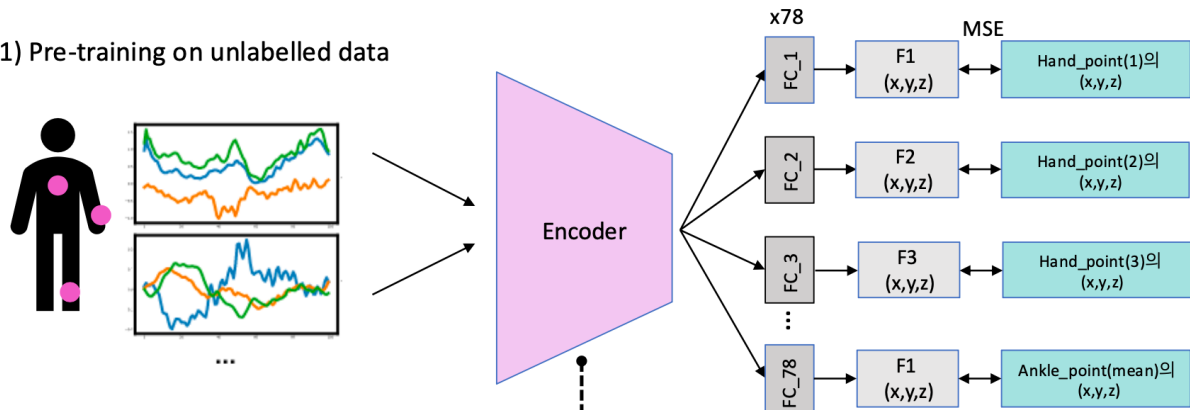
Sensor based HAR

- IMU(Inertial Measurement Unit) Sensor

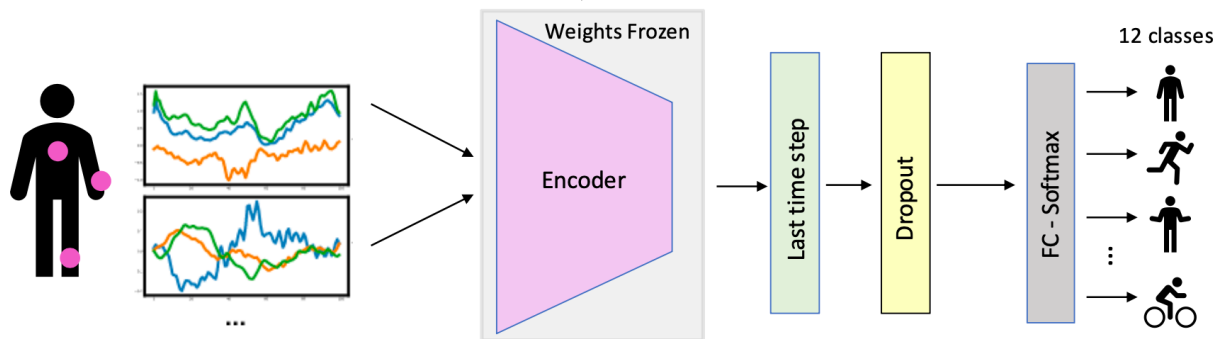


Sensor based HAR

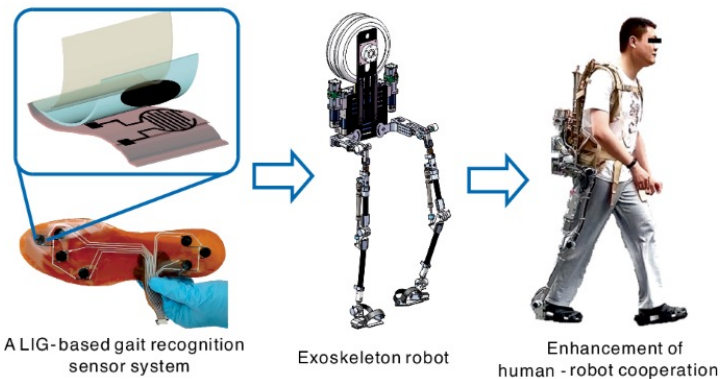
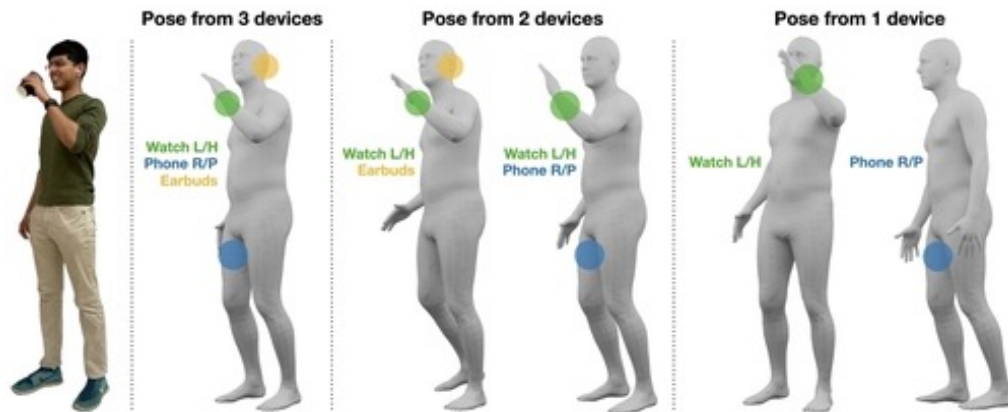
1) Pre-training on unlabelled data



2) Fine-tuning with labeled examples



HAR Application



THOHI

감사합니다!